

0.1 The selected Schmidt space

The relabeling of the selected Schmidt space in [OAI26, Section 7] is described by an explicit isometry. Let $\Omega \in \mathcal{H}_A \otimes \mathcal{H}_B$ and let $\rho_A = \text{Tr}_B |\Omega\rangle\langle\Omega|$. Choose an orthonormal eigenbasis $(u_i)_{i \in A}$ of ρ_A , with eigenvalues p_i . For a selected set $E \subseteq A$, write $z = \sum_{i \in E} p_i$ and $P_E = \sum_{i \in E} |u_i\rangle\langle u_i|$.

Definition 0.1.1 (The selected coordinate space). [Lean](#) [✓ Stmt](#) Let $\mathcal{H}_R = \mathbb{C}^E$, so that $\dim \mathcal{H}_R = |E|$. Define $J : \mathcal{H}_R \rightarrow \mathcal{H}_A$ by $J|i\rangle = u_i$.

Theorem 0.1.2 (The spectral isometry). [Lean](#) [Lean](#) [✓ Stmt](#) The selected coordinate embedding satisfies $J^\dagger J = I_R$ and $JJ^\dagger = P_E$.

Proof. [✓ Proof](#) Orthonormality gives $\langle u_i, u_j \rangle = \delta_{ij}$. The matrix $JJ^\dagger$ is $\sum_{i \in E} |u_i\rangle\langle u_i| = P_E$. $\square$

Definition 0.1.3 (The compressed selected vector). [Lean](#) [✓ Stmt](#) If $z > 0$, define $\chi = z^{-1/2}(J^\dagger \otimes I_B)\Omega$ on $\mathcal{H}_R \otimes \mathcal{H}_B$.

Theorem 0.1.4 (The marginals of the compressed selected vector). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose $z > 0$ and put $\Omega_E = z^{-1/2}(P_E \otimes I_B)\Omega$. Then $\|\chi\| = 1$ and

$$(J \otimes I_B)\chi = \Omega_E, \quad (1)$$

$$\text{Tr}_B |\chi\rangle\langle\chi| = \text{diag}_{i \in E}(p_i/z), \quad (2)$$

$$\text{Tr}_R |\chi\rangle\langle\chi| = \text{Tr}_A |\Omega_E\rangle\langle\Omega_E|. \quad (3)$$

In particular, every reduced state on a subsystem of B is preserved by this relabeling, as required in the marginal comparison of Section 7.

Proof. [✓ Proof](#) The identity $JJ^\dagger = P_E$ gives (1). The coefficient matrix of χ is $z^{-1/2}J^\dagger C_\Omega$, and hence its first marginal is $z^{-1}J^\dagger \rho_A J = \text{diag}_{i \in E}(p_i/z)$. Tracing out an isometrically embedded factor gives (3). Taking the trace gives the same squared norm as Ω_E , which is one. $\square$

0.2 A Bell projection onto the selected space

Let $\Omega \in \mathcal{H}_A \otimes \mathcal{H}_B$ and let $\rho_A = \text{Tr}_B |\Omega\rangle\langle\Omega|$. For a selected set E of eigenvectors of ρ_A , let $J : \mathbb{C}^E \rightarrow \mathcal{H}_A$ be the selected spectral isometry. Write $d = |E|$ and $z = \sum_{i \in E} p_i$, and suppose $z > 0$. The auxiliary spaces $\mathcal{H}_C$ and $\mathcal{H}_R$ are copies of $\mathbb{C}^E$. Put $u = d^{-1/2} \sum_{i \in E} |i\rangle_C |i\rangle_R$ and $\chi = z^{-1/2}(J^\dagger \otimes I_B)\Omega$. These are the one-copy constructions in [OAI26, Section 7].

Definition 0.2.1 (The selected Bell vector and its outer product). [Lean](#) [Lean](#) [✓ Stmt](#) Define

$$\beta = d^{-1/2} \sum_{i \in E} (J|i\rangle)_A \otimes |i\rangle_C, \quad (4)$$

$$P = |\beta\rangle\langle\beta|. \quad (5)$$

Theorem 0.2.2 (Normalization and orthogonal projection). [Lean](#) [Lean](#) [✓ Stmt](#) The selected Bell vector satisfies $\|\beta\| = 1$, and P is an orthogonal projection: $P^2 = P = P^\dagger$.

Proof. [✓ Proof](#) Positive selected mass implies $d > 0$. The uniform pair has squared norm $d \cdot d^{-1} = 1$. Since $J^\dagger J = I_R$, the isometry $J \otimes I_C$ preserves this norm. Thus $\langle \beta, \beta \rangle = 1$, which gives $P^2 = P = P^\dagger$. $\square$

Definition 0.2.3 (The one-copy vectors). [Lean](#) [Lean](#) [✓ Stmt](#) On $(\mathcal{H}_A \otimes \mathcal{H}_C) \otimes (\mathcal{H}_R \otimes \mathcal{H}_B)$, define $\text{pre}_{acrb} = \Omega_{ab} u_{cr}$ and $\text{post} = \beta \otimes \chi$.

Theorem 0.2.4 (The one-copy Bell contraction). [Lean](#) [✓ Stmt](#) For every Ω with positive selected mass,

$$(P \otimes I_{RB})\text{pre} = \frac{\sqrt{z}}{d} \text{post}. \quad (6)$$

Proof. [✓ Proof](#) Let C_Ω be the coefficient matrix of Ω . The Kronecker delta in the auxiliary uniform pair gives

$$((\langle \beta|_{AC} \otimes I_{RB})\text{pre})_{rb} = d^{-1}(J^\dagger C_\Omega)_{rb} = \frac{\sqrt{z}}{d} \chi_{rb}. \quad (7)$$

Multiplication by $|\beta\rangle$ gives (6). $\square$

0.3 Bell projections on finitely many copies

Let $\Omega \in \mathcal{H}_A \otimes \mathcal{H}_B$, and select a set E of eigenvectors of its actual marginal $\rho_A = \text{Tr}_B |\Omega\rangle\langle\Omega|$. Write $d = |E|$ and $z = \sum_{i \in E} p_i$, and suppose $z > 0$. Let β , $P = |\beta\rangle\langle\beta|$, pre and $\text{post} = \beta \otimes \chi$ be the one-copy constructions of Section 0.2. Set $c = \sqrt{z}/d$. For each $k \geq 0$, tensor powers are written in the basis indexed by k -tuples: the coefficient of $v^{\otimes k}$ at x is $\prod_{j=0}^{k-1} v(x_j)$. These are the finite-copy consequences of [OAI26, Section 7].

Theorem 0.3.1 (The finite-copy Bell contraction). [Lean](#) [✓ Stmt](#) For every $k \geq 0$,

$$(P \otimes I_{RB})^{\otimes k} \text{pre}^{\otimes k} = c^k \text{post}^{\otimes k}. \quad (8)$$

The vector Ω need not have unit norm.

Proof. [✓ Proof](#) For finite families of matrices and vectors, distributivity gives

$$\sum_y \prod_j M_j(x_j, y_j) v_j(y_j) = \prod_j \sum_{y_j} M_j(x_j, y_j) v_j(y_j). \quad (9)$$

Apply the one-copy Bell contraction to each factor. The k scalar factors give $\prod_j c = c^k$. $\square$

Theorem 0.3.2 (The repeated Bell vector and its projection). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $k \geq 0$, $\|\beta^{\otimes k}\| = 1$, and $P^{\otimes k}$ is an orthogonal projection. Simultaneously permuting the AC copies fixes $\beta^{\otimes k}$. The permutation assertion holds without the assumption $z > 0$.

Proof. [✓ Proof](#) The same finite-product expansion gives $\langle \beta^{\otimes k}, \beta^{\otimes k} \rangle = \prod_j \langle \beta, \beta \rangle = 1$. Entrywise multiplication identifies $P^{\otimes k} = |\beta^{\otimes k}\rangle\langle\beta^{\otimes k}|$, so it is self-adjoint and idempotent. For any permutation σ , reindexing the finite product gives $\prod_j \beta(x_{\sigma(j)}) = \prod_j \beta(x_j)$. $\square$

Theorem 0.3.3 (An operator on disjoint auxiliary copies). [Lean](#) [✓ Stmt](#) Regroup the copies canonically as $\mathcal{H}_{AC}^{\otimes k} \otimes (\mathcal{H}_R^{\otimes k} \otimes \mathcal{H}_B^{\otimes k})$. Denote the transported product vectors by pre_k and post_k . For any operator Λ on $\mathcal{H}_R^{\otimes k}$, put $Q = P^{\otimes k} \otimes I_{R^k B^k}$ and $T_\Lambda = I_{(AC)^k} \otimes (\Lambda \otimes I_{B^k})$. Then

$$QT_\Lambda \text{pre}_k = c^k T_\Lambda \text{post}_k. \quad (10)$$

No self-adjointness, projection property or commutation assumption is required for Λ .

Proof. ✓ Proof The coordinate equivalence acts by

$$((a_j, c_j), (r_j, b_j))_j \mapsto ((a_j, c_j)_j, ((r_j)_j, (b_j)_j)). \quad (11)$$

It transports $(P \otimes I_{RB})^{\otimes k}$ to Q . Kronecker multiplication gives

$$QT_\Lambda = P^{\otimes k} \otimes (\Lambda \otimes I_{B^k}) = T_\Lambda Q. \quad (12)$$

Transport (8) and apply T_Λ . Using (12) gives (10). □

0.4 Reduced densities and masses of repeated pure states

Let $\mathcal{H}_S = \mathbb{C}^S$ and $\mathcal{H}_R = \mathbb{C}^\Omega$ be finite-dimensional, and let $\psi \in \mathcal{H}_S \otimes \mathcal{H}_R$. Write $\rho = \text{tr}_S |\psi\rangle\langle\psi|$ for its actual reduced density, without imposing normalization. For $k \geq 0$, regroup the repeated vector into the two systems. Its coordinates, indexed by $x \in S^k$ and $y \in \Omega^k$, are

$$v_k(x, y) = \prod_{i=0}^{k-1} \psi(x_i, y_i). \quad (13)$$

These are the pure-state identities used in the Schur-sector argument of [OAI26, Section 7].

Theorem 0.4.1 (Reduced density of a repeated bipartite vector). Lean ✓ Stmt *For every ψ and every $k \geq 0$,*

$$\text{tr}_{S^k} |v_k\rangle\langle v_k| = \rho^{\otimes k}. \quad (14)$$

Proof. ✓ Proof For $y, z \in \Omega^k$, the matrix entry on the left is

$$\sum_{x \in S^k} \prod_i \psi(x_i, y_i) \overline{\psi(x_i, z_i)} = \prod_i \sum_{s \in S} \psi(s, y_i) \overline{\psi(s, z_i)} = \prod_i \rho(y_i, z_i). \quad (15)$$

The middle equality is the distributive law for a finite product of sums. □

Theorem 0.4.2 (Projection mass of a repeated bipartite vector). Lean ✓ Stmt *For every orthogonal projector P on $\mathcal{H}_R^{\otimes k}$,*

$$\|(I_{S^k} \otimes P)v_k\|^2 = \text{Re tr}(\rho^{\otimes k} P). \quad (16)$$

In particular, taking P to be an actual central Schur projector gives the squared norm in that sector.

Proof. ✓ Proof The trace-pairing property of the partial trace and the preceding identity give

$$\text{tr}(\rho^{\otimes k} P) = \langle v_k, (I_{S^k} \otimes P)v_k \rangle = \langle (I_{S^k} \otimes P)v_k, (I_{S^k} \otimes P)v_k \rangle. \quad (17)$$

The second equality uses $P = P^\dagger = P^2$. Taking real parts proves the formula. □

Both identities include zero copies and empty one-copy factors; no invertibility of ρ is required. Concentration, the entropy window and selection of a common sequence of labels are separate assertions.

0.5 A common sequence of high-mass entropy labels

Let $q = \dim \mathcal{H}_R$ and $N_k = (k+1)^{q^2}$. The permutation action on $\mathcal{H}_R^{\otimes k}$ has central projectors Π_λ , indexed by irreducible labels λ of the symmetric group. Write d_λ for the irreducible dimension. The following statements give the simultaneous selection in [OAI26, Section 7], equation (comparator:high-label).

Theorem 0.5.1 (Finite entropy-window criterion). Lean ✓ Stmt *Let σ be a density matrix on $\mathcal{H}_R^{\otimes k}$ which commutes with all copy permutations. Let $a \in \mathbb{R}$ and $w \geq 0$. Set $L = -\log \sigma$, with the logarithm assigned value zero on the kernel. If*

$$\mathrm{tr}(\sigma 1_{\{|L-a|>w\}}) + N_k e^{-w/2} \leq \frac{1}{2}, \quad (18)$$

then there is a label λ such that

$$\mathrm{tr}(\sigma \Pi_\lambda) \geq \frac{1}{2N_k}, \quad |\log d_\lambda - a| \leq 2w. \quad (19)$$

Proof. ✓ Proof Resolve σ and the central label projectors simultaneously. On a joint eigenspace of positive density eigenvalue u and label λ , put $\ell = -\log u$ and $f = \log d_\lambda$. The eigenvalue multiplicity bound $ud_\lambda \leq 1$ implies $f \leq \ell$. If $|f - a| > 2w$, then either $|\ell - a| > w$ or $\ell - f > w$. The mass of the second event is bounded by $e^{-w/2}$ times the exponential moment of $(L - F)/2$, which is at most N_k . The hypothesis therefore leaves at least half of the total mass in the good labels. At most N_k labels occur, so one good label has mass at least $(2N_k)^{-1}$. Zero eigenvalues contribute no mass. $\square$

Theorem 0.5.2 (Independent-copy entropy window). Lean ✓ Stmt *For every density matrix ρ on $\mathcal{H}_R$, for all sufficiently large k there is a label λ satisfying*

$$\mathrm{tr}(\rho^{\otimes k} \Pi_\lambda) \geq \frac{1}{2N_k}, \quad |\log d_\lambda - kS(\rho)| \leq 2k^{3/4}. \quad (20)$$

No invertibility of ρ is assumed.

Proof. ✓ Proof The independent-copy surprisal has mean $kS(\rho)$ and variance kV , where V is its one-copy variance. Chebyshev's inequality at width $k^{3/4}$ bounds the tail by $V/\sqrt{k}$. The sum $V/\sqrt{k} + N_k e^{-k^{3/4}/2}$ tends to zero. The finite criterion applies for all sufficiently large k . $\square$

Theorem 0.5.3 (One sequence for mass and entropy). Lean ✓ Stmt *For every density matrix ρ , there is one sequence $(\lambda_k)_{k \geq 0}$ such that, for all sufficiently large k ,*

$$\mathrm{tr}(\rho^{\otimes k} \Pi_{\lambda_k}) \geq \frac{1}{2N_k}, \quad (21)$$

and

$$\log d_{\lambda_k} = kS(\rho) + o(k). \quad (22)$$

Proof. ✓ Proof Choose a label satisfying both conclusions of the preceding theorem for each sufficiently large k ; at the remaining finitely many values choose any label. The normalized dimension error is at most $2k^{-1/4}$ and therefore tends to zero. $\square$

Theorem 0.5.4 (High-mass labels of actual pure tensor powers). [Lean](#) [✓ Stmt](#) Let $\psi \in \mathcal{H}_S \otimes \mathcal{H}_R$ be a unit vector, let $\rho = \text{tr}_S |\psi\rangle\langle\psi|$, and let v_k be its literal tensor power regrouped into $\mathcal{H}_S^{\otimes k} \otimes \mathcal{H}_R^{\otimes k}$. There is one sequence (λ_k) for which, for all sufficiently large k ,

$$\|(I \otimes \Pi_{\lambda_k})v_k\|^2 \geq \frac{1}{2N_k}, \quad (23)$$

and $\log d_{\lambda_k} = kS(\rho) + o(k)$.

Proof. [✓ Proof](#) The actual reduced density is positive semidefinite and has trace one. Apply the common-sequence theorem to this density. The projection-mass identity for actual pure tensor powers converts its trace mass into the stated squared norm. $\square$

0.6 The initial vector with common auxiliary labels

Let $\Omega \in \mathbb{C}^A$ and $\psi \in \mathbb{C}^S \otimes \mathbb{C}^d$ be unit vectors. Suppose that $H\Omega = E_0\Omega$, with $E_0 \in \mathbb{R}$. Write $\rho = \text{tr}_S |\psi\rangle\langle\psi|$. The initial vector below is the construction in [OAI26, Section 7], equations (comparator:prevector) and (comparator:high-label).

Definition 0.6.1 (Initial vector in product coordinates). [Lean](#) [✓ Stmt](#) Let $u_d = d^{-1/2} \sum_{i=0}^{d-1} |i, i\rangle$ be the uniform auxiliary pair. Regroup its literal tensor power into the two auxiliary systems and write it as U_k . For a central label projector Π_λ on $(\mathbb{C}^d)^{\otimes k}$, define

$$v_{k,\lambda} = \Omega^{\otimes k} \otimes (I_C \otimes \Pi_\lambda)U_k. \quad (24)$$

All products and projections here act on the actual finite copy spaces.

Theorem 0.6.2 (One sequence for the initial vector and selected mass). [Lean](#) [✓ Stmt](#) There is one sequence (λ_k) for which every $k \geq 1$ satisfies

$$0 < \|v_{k,\lambda_k}\| \leq 1, \quad (\overline{H}_k \otimes I_{CR})v_{k,\lambda_k} = E_0 v_{k,\lambda_k}, \quad \overline{H}_k = \frac{1}{k} \sum_{i=0}^{k-1} H^{(i)}. \quad (25)$$

The vector has label λ_k on both auxiliary systems and is invariant under every simultaneous copy permutation on the physical, left auxiliary and right auxiliary systems. For all sufficiently large k ,

$$\|(I_{S^k} \otimes \Pi_{\lambda_k})\psi^{\otimes k}\|^2 \geq \frac{1}{2(k+1)^{d^2}}, \quad (26)$$

and $\log d_{\lambda_k} = kS(\rho) + o(k)$. The initial projection of U_k is required only to be nonzero; no inverse polynomial lower bound on its probability is asserted.

Proof. [✓ Proof](#) Start with the common sequence selected from the actual density ρ . For any remaining copy number at which the selected sector has zero mass, choose a sector of positive mass using the complete central resolution of $\rho^{\otimes k}$. The original sequence has positive mass eventually, so these changes affect only finitely many terms and preserve the mass bound and entropy asymptotic.

Unit normalization of ψ implies $d > 0$. Every selected projector is nonzero, and hence acts nontrivially on the uniform pair. Tensoring with $\Omega^{\otimes k}$ preserves its norm. Orthogonal projection is contractive, and both unprojected tensor powers have norm one, giving the asserted norm

bounds. The one-copy eigenvector equation gives the mean-energy equation on the physical product, irrespective of the auxiliary vector.

The repeated uniform pair has matching left and right central labels. Idempotence therefore gives both fixed-label equations after projection. The uniform pair and the physical product are invariant under simultaneous copy permutations; centrality of the right label projector preserves this invariance after projection. $\square$

0.7 Schmidt truncation and the common-label Bell projection

Let $\Omega \in \mathcal{H}_A \otimes \mathcal{H}_B$, and choose an orthonormal eigenbasis $(u_i)_{i \in A}$ of its actual marginal $\rho_A = \text{Tr}_B |\Omega\rangle\langle\Omega|$, with eigenvalues p_i . Fix $E \subseteq A$, put $d = |E|$ and $z = \sum_{i \in E} p_i$, and suppose $z > 0$. Identify E with $\{0, \dots, d-1\}$ by a bijection ι . In these coordinates, set

$$J|r\rangle = u_{\iota^{-1}(r)}, \quad \chi = z^{-1/2}(J^\dagger \otimes I_B)\Omega, \quad \beta = \frac{1}{\sqrt{d}} \sum_{r=0}^{d-1} J|r\rangle \otimes |r\rangle. \quad (27)$$

The same coordinates are used on the two auxiliary systems C and R . These constructions instantiate the Schmidt truncation and pinning step in [OAI26, Section 7], equations (comparator:prevector), (comparator:high-label) and (comparator:bell-pin). The inverse-metric comparator bounds require the subsequent geometric and energy estimates.

Theorem 0.7.1 (The actual initial vector under the selected Bell projection). [Lean](#) [Stmnt](#) For $k \geq 0$ and any actual central label λ on R^k , let

$$v_{k,\lambda} = \Omega^{\otimes k} \otimes (I_C \otimes \Pi_\lambda) u_d^{\otimes k}, \quad u_d = \frac{1}{\sqrt{d}} \sum_{r=0}^{d-1} |r, r\rangle. \quad (28)$$

Regroup the factors into $(AC)^k \otimes (R^k \otimes B^k)$ and put $Q_k = (|\beta\rangle\langle\beta|)^{\otimes k} \otimes I_{R^k B^k}$. Then

$$Q_k v_{k,\lambda} = \left(\frac{\sqrt{z}}{d} \right)^k [I_{(AC)^k} \otimes (\Pi_\lambda \otimes I_{B^k})] (\beta^{\otimes k} \otimes \chi^{\otimes k}). \quad (29)$$

The right label acts on a factor disjoint from the Bell projection.

Proof. [Stmnt](#) Transport the selected coordinates from E to $\{0, \dots, d-1\}$. This transports the selected Bell projection, compressed vector and uniform pair by the same bijection. The finite-copy Bell identity for a disjoint operator therefore applies to the actual projector Π_λ . The coordinate regrouping is

$$((a_j, c_j)_j, ((r_j)_j, (b_j)_j)) \mapsto ((a_j, b_j)_j, ((c_j)_j, (r_j)_j)). \quad (30)$$

Matrix vectorization shows that the projected input is precisely (28): its physical factor is unchanged by the right auxiliary projection. Applying the finite-copy identity gives (29), with all k scalar factors equal to $\sqrt{z}/d$. $\square$

Theorem 0.7.2 (The common sequence selected from the actual Schmidt truncation). [Lean](#) [Stmnt](#) Suppose in addition that $\|\Omega\| = 1$ and $H\Omega = E_0\Omega$, where $E_0 \in \mathbb{R}$. Write $\psi(b, r) = \chi(r, b)$, so that

$$\|\psi\| = 1, \quad \rho_R = \text{Tr}_B |\psi\rangle\langle\psi| = \text{diag}_{0 \leq r < d} (p_{\iota^{-1}(r)}/z). \quad (31)$$

There is one sequence (λ_k) such that, for every $k \geq 1$, the actual initial vector in (28) is nonzero, has norm at most one, and satisfies

$$(\overline{H}_k \otimes I_{CR})v_{k,\lambda_k} = E_0 v_{k,\lambda_k}, \quad \overline{H}_k = \frac{1}{k} \sum_{j=0}^{k-1} H^{(j)}. \quad (32)$$

It has label λ_k on both C and R , and is fixed by simultaneous copy permutations of the physical and both auxiliary systems. For all sufficiently large k ,

$$\|(I_{B^k} \otimes \Pi_{\lambda_k})\psi^{\otimes k}\|^2 \geq \frac{1}{2(k+1)^{d^2}}, \quad \log d_{\lambda_k} = kS(\rho_R) + o(k). \quad (33)$$

The Bell identity (29) applies to this same sequence.

Proof. ✓ Proof The compressed-vector marginal identity, followed by the coordinate bijection and the interchange of the two factors, gives (31). These coordinate changes preserve the unit norm. Apply the common-sequence theorem for the actual initial vector to this ψ , the physical vector Ω , and its eigenvector equation. Its selected mass and entropy are those of the actual density ρ_R in (31). $\square$

Bibliography

- [OAI26] OpenAI, *A two-dimensional area law from a global spectral gap*, September 24, 2026, Section 7.